ALGEBRA 1 · EXPONENTS AND SCIENTIFIC NOTATION

Exponent Rules Unpacker

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Exponent Rules Unpacker** on Mr. K's Classroom: Algebra 1 → Exponents and Scientific Notation → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. Predict: $x^3 \cdot x^4 = x^{?}$

   x^7

2. Predict the value of 2^0.

   1

2 Explore use the tool and record what you see

Unpack the powers

Use the product, quotient and power rule tabs.

Expression	Written as factors	Number of x's left	Simplified
$x^2 \cdot x^3$	**x·x · x·x·x**	**5**	**x^5**
$x^4 \cdot x$	**x·x·x·x · x**	**5**	**x^5**
$\frac{x^5}{x^2}$	**2 pairs cancel**	**3 on top**	**x^3**
$\frac{x^2}{x^5}$	**2 pairs cancel**	**3 on the bottom**	**$\frac{1}{x^3} = x^{-3}$**
$(x^2)^3$	**3 groups of x·x**	**6**	**x^6**

Zero and negative exponents (base 2)

Each step down divides by 2.

Power	2^3	2^2	2^1	2^0	2^{-1}	2^{-2}
Value	8	**4**	**2**	**1**	**$\frac{1}{2}$**	**$\frac{1}{4}$**

3 Write the rule fill in each blank from what you observed

1. When you multiply powers with the same base, you **add** the exponents.
2. When you divide powers with the same base, you **subtract** the exponents.
3. When you raise a power to a power, you **multiply** the exponents.
4. Any nonzero number to the zero power equals **1**.
5. x^{-n} = **$\frac{1}{x^n}$** .

4 Check no tool: show your work

1. Simplify $a^6 \cdot a^2$.

   a^8

2. Simplify $\frac{y^9}{y^4}$.

   y^5

3. Simplify $(m^3)^4$.

   m^{12}

4. Evaluate 5^{-2}.

   $\frac{1}{25}$

5. Simplify $\frac{x^3}{x^3}$ two ways.

   x^0 and 1, so $x^0 = 1$

5 Reflect

Use the pattern table to explain why $3^0 = 1$.

Each step down divides by 3: $3^1 = 3$, so $3^0 = 3 \div 3 = 1$.